Stakeholder Requirements.

1. End-User Requirements (Field Technicians & Hardware Enthusiasts)

Directional Audio Isolation: The system must accurately pinpoint the 3D direction of specific sounds like, a single conversation or a clap and effectively mute or isolate them from background noise.

On-Site Data Visualization: The device must generate a 2D time-by-frequency spectrogram, allowing technicians to view and analyze the data directly in the field without needing to transfer it to a separate workstation.

Directional Steering: Users must have the ability to manually or digitally steer the directional focus of the microphone, with real-time or near-real-time reflection of these changes on the spectrogram via beamforming.

Compact Form Factor: The microphone and boom must be compact enough to be placed into tight or awkward spaces where a user cannot easily fit their head.

Control Interface: The steering controls must be intuitive to use. If digital, the system should support saving and restoring specific state configurations.

2. Environmental & Durability Requirements

Ruggedization: The microphone array, boom, and casing must be highly durable, capable of withstanding accidental drops and the rigors of standard field use.

Moisture Resistance: The device must continue to function effectively when exposed to mild moisture or damp conditions.

3. Data & Export Requirements

Data Portability: The system must allow users to export the captured audio and directional data (like Time Difference of Arrival - TDOA) for post-capture processing and deeper analysis on external software.

4. Manufacturing & Engineering Stakeholders

Component Sourcing: The system design should prioritize the use of industry-standard components (PCBs, mics) and established wave computation pipelines (like the MUSIC algorithm) to ensure it is cost-effective to manufacture and assemble.

Testing Accessibility: The design must allow for variable microphone distances to easily test and calibrate alternate sound calculation pipelines during the prototyping phase.